

Contents

1	Introduction	1
2	Examples of Reflexive Spaces	2
3	Properties of Reflexive Spaces	2

1 Introduction

Let X be a normed vector space; we let X' denote the dual space of X and X'' denote the dual space of X' . For each $x \in X$ we define the “evaluation functional” $\text{ev}_x : X' \rightarrow \mathbf{R}$ which maps $x' \mapsto x'(x)$. These are linear since $(x' + y')(x) = x'(x) + y'(x)$ and $(\alpha x')(x) = \alpha x'(x)$ by definition of the addition and scalar multiplication on X' . We define the map $E : X \rightarrow X''$ by $x \mapsto \text{ev}_x$; we have the following:

Proposition 1

The map E has the following properties:

1. E is linear;
2. E is injective;
3. E is isometric i.e. $\|\text{ev}_x\|_{X''} = \|x\|_X$ (and thus $\text{ev}_x \in X''$).

Proof. We first show linearity: let $x, y \in X$; for any $x' \in X'$ we have

$$\text{ev}_x(x') + \text{ev}_y(x') = x'(x) + x'(y) = x'(x + y) = \text{ev}_{x+y}(x')$$

so $\text{ev}_x + \text{ev}_y = \text{ev}_{x+y}$. Similarly for $x \in X$ and $\alpha \in \mathbf{R}$ we have

$$\alpha \text{ev}_x(x') = \alpha x'(x) = x'(\alpha x) = \text{ev}_{\alpha x}(x')$$

so $\text{ev}_{\alpha x} = \alpha \text{ev}_x$.

Now suppose $x \in X$ such that $x \neq 0$ and define the linear functional $f : \text{Span}\{x\} \rightarrow \mathbf{R}$ by $f(\alpha x) = \alpha$. This is bounded, so by the Hahn-Banach theorem there exists a functional $F \in X'$ which extends f . By construction, $\text{ev}_x(F) = F(x) = f(x) = 1$ so ev_x is not the zero functional. Thus $N(E) = \{0\}$ so E is injective.

Finally, let $x \in X$; if $x = 0$ then ev_x is the zero map so $\|x\|_X = \|\text{ev}_x\|_{X''} = 0$. So suppose $x \neq 0$ and let $x' \in X'$ with $x' \neq 0$. Then

$$|\text{ev}_x(x')| = |x'(x)| \leq \|x'\|_{X'} \|x\|_X$$

so $\|\text{ev}_x\|_{X''} \leq \|x\|_X$. We will now construct $y' \in X'$ such that $|\text{ev}_x(x')| = \|y'\|_{X'} \|x\|_X$. Define $f : \text{Span}\{x\} \rightarrow \mathbf{R}$ by $f(\alpha x) = \alpha$. Then $|f(\alpha x)| = \frac{|\alpha| \|x\|_X}{\|x\|_X} = \frac{\|\alpha x\|_X}{\|x\|_X}$ so $\|f\|_{\text{op}} = \frac{1}{\|x\|_X}$. By the Hahn-Banach theorem, there exists $y' \in X'$ which extends f such that $\|y'\|_{X'} = \|f\|_{\text{op}} = \frac{1}{\|x\|_X}$. By construction,

$$|\text{ev}_x(y')| = |y'(x)| = |f(x)| = 1 = \frac{\|x\|_X}{\|x\|_X} = \|x\|_X \|y'\|_{X'}$$

so $\|\text{ev}_x\| = \|x\|_X$ as claimed. ■

With the above, we can make the following definition:

Reflexive Spaces

For X a normed vector space, we say that X is *reflexive* if the map $E : X \rightarrow X''$ defined by $x \mapsto \text{ev}_x$ is surjective (equivalently, if E is an isometric isomorphism).

2 Examples of Reflexive Spaces

We can immediately prove some examples of spaces that are reflexive:

Proposition 2

Every finite dimensional space is reflexive.

Proof. Suppose X is a finite dimensional space and let $\{e_1, \dots, e_n\}$ be a basis for X . The dual basis $\{e^1, \dots, e^n\}$ defined by $e^i(e_j) = \delta_j^i$ (where δ_j^i is the Kronecker delta) forms a basis for X' , so X' has the same dimension as X . From this it follows that X'' has the same dimension as X , so E is surjective since it is injective. ■

Proposition 3

Every Hilbert space is reflexive.

Proof. Suppose X is a Hilbert space; by the Riesz representation theorem, every $x' \in X'$ can be considered as the inner product $\langle \cdot, x \rangle$ for some $x \in X$. With this we can consider X' as a Hilbert space with the inner product

$$\langle x', y' \rangle_{X'} = \langle x, y \rangle_X$$

where x, y are such that $x'(z) = \langle z, x \rangle$ and $y'(z) = \langle z, y \rangle$. Then for any $x'' \in X''$, we can use the Riesz Representation Theorem to find a $x' \in X'$ such that $x''(z') = \langle z', x' \rangle_{X'}$ for all $z' \in X'$. However, we have $\langle z', x' \rangle_{X'} = \langle z, x \rangle_X = \langle x, z \rangle_X = z'(x)$ so X is reflexive. ■

Proposition 4

l_p is reflexive for every $1 < p < \infty$.

Proof. Let $q \in (1, \infty)$ so that $pq = p + q$; for each $x' \in l'_p$ there exists $x \in l_q$ such that

$$x'(z) = \sum_{k=1}^{\infty} z_k x_k$$

for all $z \in l_p$. Now let $x'' \in l''_p$; identifying l'_p with l_q , we can find $x \in l_p$ such that $x''(z') = \sum_{k=1}^{\infty} z_k x_k$ for all $z' \in l_q$. However,

$$\sum_{k=1}^{\infty} z_k x_k = \sum_{k=1}^{\infty} x_k z_k = z'(x).$$

3 Properties of Reflexive Spaces

The natural question to ask is what special properties do we get in reflexive spaces? The following begins to answer both these questions:

Theorem 1

All reflexive spaces are Banach spaces.

Proof. Suppose X is reflexive; then E is an isometric isomorphism to X'' . We know that the dual of any normed vector space is complete, so we have that X'' is a Banach space. Now let (x_n) be a Cauchy sequence in X ; we note that

$$\|Ex_n - Ex_m\|_{X''} = \|E(x_n - x_m)\|_{X''} = \|x_n - x_m\|_X$$

so the sequence (Ex_n) is Cauchy in X'' . By completeness of X'' we have that $Ex_n \rightarrow z$ for some $z \in X''$; we claim that $x_n \rightarrow E^{-1}z$ in X . Indeed, we have

$$\|x_n - E^{-1}z\|_X = \|E(x_n - E^{-1}z)\|_{X''} = \|Ex_n - EE^{-1}z\|_{X''} = \|Ex_n - z\|_{X''} \rightarrow 0$$

so X is complete as claimed. ■

Proposition 5

Let X be a reflexive space and $Z \subset X$ a closed subspace. Then Z is also reflexive.

Proof. Since X is reflexive, X is a Banach space by **theorem 1**. As a closed subspace of a Banach space, Z is also a Banach space. For any $x' \in X'$ we can define an element of Z' by considering the restriction $x'|_Z$; we note that $\|x'|_Z\|_{Z'} \leq \|x'\|_{X'}$ as the supremum is taken over a smaller set. Let $z'' \in Z''$; we construct an element $x'' \in X''$ by $x''(x') = z''(x'|_Z)$. Then x'' is linear and

$$|x''(x')| = |z''(x'|_Z)| \leq \|z''\|_{Z''} \|x'|_Z\|_{Z'} \leq \|z''\|_{Z''} \|x'\|_{X'}$$

so $x'' \in X''$. Since X is reflexive, there exists $x \in X$ such that $x'' = \text{ev}_x$; it now follows that $z''(x'|_Z) = x'(x)$ for all $x' \in X'$. By the Hahn-Banach theorem, every $z' \in Z'$ can be considered as $x'|_Z$ for some $x' \in X'$ so it suffices to show that $x \in Z$. For any $y' \in Z^\circ$, we have that $y'|_Z = 0$ so $y'(x) = z''(y'|_Z) = z''(0) = 0$. This means that $x \in {}^\circ(Z^\circ)$, but Z is closed so by lemma 3.5 in [1], we have ${}^\circ(Z^\circ) = Z$, meaning $x \in Z$ as desired. ■

Theorem 2

Let X be a Banach space. Then X is reflexive if and only if X' is reflexive.

Proof. Suppose X is reflexive and let $x''' \in X'''$; it follows that $x \mapsto x'''(Ex)$ is a bounded linear functional on X . Thus there exists $x' \in X'$ such that $x'''(Ex) = x'(x)$ for all $x \in X$. However, $x'(x) = \text{ev}_x(x')$ by definition so $x'''(Ex) = \text{ev}_x(x')$. Since X is reflexive, every $x'' \in X''$ can be represented as ev_x for some $x \in X$. Thus we have that $x'''(x'') = x''(x')$ for all $x'' \in X''$. This means we can write $x''' = \text{ev}_{x'}$ so X' is reflexive.

Suppose now that X' is reflexive; we first note that $R(E)$ is a closed subspace of X'' . To see this, let ev_{x_n} be a sequence in $R(E)$ which converges to $x'' \in X''$. Then since E is an isometry, x_n is a Cauchy sequence in $R(E)$; this implies that $x_n \rightarrow x$ in X so $\text{ev}_{x_n} \rightarrow \text{ev}_x$ in X'' by continuity of E . Thus $x'' = \text{ev}_x$ by uniqueness of limits so $R(E)$ is closed. Let $x''' \in R(E)^\circ$; since X' is reflexive there exists $x' \in X'$ such that $x''' = \text{ev}_{x'}$. Then for all $x \in X$,

$$x'(x) = \text{ev}_x(x') = \text{ev}_{x'}(\text{ev}_x) = x'''(\text{ev}_x) = 0$$

so $x' = 0$. This implies that $\text{ev}_{x'} = 0$ by linearity, so $R(J)^\circ = \{0\}$ meaning $R(J) = X''$. ■

The previous theorem in particular allows us to show that some spaces are not reflexive; we know that $c'_0 = l_1$ and $l'_1 = l_\infty$ so c_0 is not reflexive. This then implies that l_1 as the dual of c_0 and l_∞ as the dual of l_1 are not reflexive either.

References

- [1] Martin Schechter. *Principles of Functional Analysis: Second Edition*. American Mathematical Society, 2002.